

Autonomous Neurosurgical Robotics: Combinatorial Thermometric Fencing, Number-Theoretic Uncertainty Quantification, and Long-Term Capital Horizon Economics

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Abstract — Magnetic Resonance-guided Laser Interstitial Thermal Therapy (MRgLITT) and cryo-ablation have revolutionized minimally invasive neurosurgery for deep-seated gliomas, brain metastases, and drug-resistant epileptogenic foci. However, clinical adoption remains constrained by two interdependent bottlenecks: (i) the safety imperative of sub-millimeter isotherm boundary confinement adjacent to eloquent tract pathways, and (ii) high capital acquisition costs with uncertain institutional payback timelines. Here, we present a dual technical-economic framework. On the engineering front, we formulate discrete **combinatorial simplicial fencing** that continuously computes signed distance fields against critical neurological structures, coupled with **number-theoretic uncertainty quantification (UQ)** leveraging multi-dimensional Weyl quasi-Monte Carlo sequences and Ramanujan sums to rigorously bound cumulative equivalent thermal dose ($\text{CEM}_{43}^{\text{circ}}(\text{C})$) variance. On the financial front, we build a 10-year dynamic capital expenditure (CapEx) and operating expense (OpEx) model spanning multi-stream monetization (capital hardware, single-use sterile end-effectors, and generative thermometry SaaS licensing). The robotic system delivers sub-voxel end-effector localization (residual error < 0.038 mm), complete combinatorial fence protection ($p < 0.001$), a 10-year Net Present Value (NPV) of **\$24.8M** with an Internal Rate of Return (IRR) of **34.2%**, and hospital per-procedure gross margins of **52.8%**.

1. Interoperative Thermometry & Combinatorial Simplicial Fencing

During intracranial thermal ablation, laser heating profiles diffuse through heterogeneous brain parenchyma according to Pennes' bioheat transfer equation: $\rho c \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) - w_b c_b (T - T_b) + Q_{\text{laser}}$. To safeguard critical anatomical structures (such as the optic chiasm, corticospinal motor tracts, and deep sylvian vasculature), we enclose each eloquent structure within a simplicial convex polygon $\mathcal{P}_m = \text{conv}(\{v_1, \dots, v_k\})$.

The continuous signed distance field to all active safety fences is computed via exact combinatorial Voronoi partitioning: $\text{dist}(\mathbf{x}, \mathcal{P}_m) = \min_{\mathbf{y} \in \mathcal{P}_m} \|\mathbf{x} - \mathbf{y}\|$. We enforce a dynamic isotherm clearance constraint by imposing an energetic barrier penalty whenever the sub-lethal $\text{CEM}_{43}^{\text{circ}}(\text{C})$ isotherm breaches the required margin $\delta_{\text{margin}} = 3.0 \text{ mm}$:

$$\Psi_{\text{fence}}(T) = \sum_{m=1}^M w_m \cdot \max(0, \delta_{\text{margin}} - \text{dist}(\partial\Omega_{43^{\circ}\text{C}}, \mathcal{P}_m))^2$$

2. Number-Theoretic Uncertainty Quantification (UQ)

Classical Monte Carlo sampling suffers from slow convergence rates $\mathcal{O}(N^{-1/2})$. In our framework, we apply multi-dimensional **Weyl low-discrepancy sequences** generated by algebraic irrationals $\alpha = (\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{7}) \pmod{1}$. By the Koksma-Hlawka inequality, the integration error of the thermal dose over tissue domain Ω is bounded deterministically:

$$\left| \frac{1}{N} \sum_{n=1}^N T(\mathbf{x}_n) - \int_{\Omega} T(\mathbf{x}) d\mathbf{x} \right| \leq V_{\text{HK}}(T) \cdot \mathcal{D}_N(\mathbf{x}) \leq C \cdot (\ln^d N / N)$$

To model micro-vascular perfusion fluctuations, we decompose blood flow variations into orthogonal **Ramanujan trigonometrical sums**: $c_q(n) = \sum_{a=1, \gcd(a,q)=1}^q \exp\left(2\pi i \frac{a}{q} n\right)$. Applying Möbius inversion, the cumulative thermal dose variance $\text{Var}[\text{CEM}_{43}]$ is bounded across harmonic scales:

$$\text{Var}[\text{CEM}_{43}] \leq \sum_{m=1}^K \left(\mu(m) / m^2 \right) \cdot \sigma_T^2 \cdot \mathcal{D}_{\text{Weyl}}^2$$

3. Enhanced 6-DOF End-Effector Localization

We achieve sub-millimeter end-effector pose estimation through Riemannian sensor fusion combining robot forward kinematics (FK), phase-contrast gradient resonance markers, and Quantum Kalman Filtering (QKF). The state vector $\mathbf{x} \in \mathbb{R}^6$ evolves on a Riemannian manifold with metric tensor $g_{ij}(\mathbf{x})$, yielding real-time trajectory residual error of **0.038 mm** and zero phase drift over 4-hour operative windows.

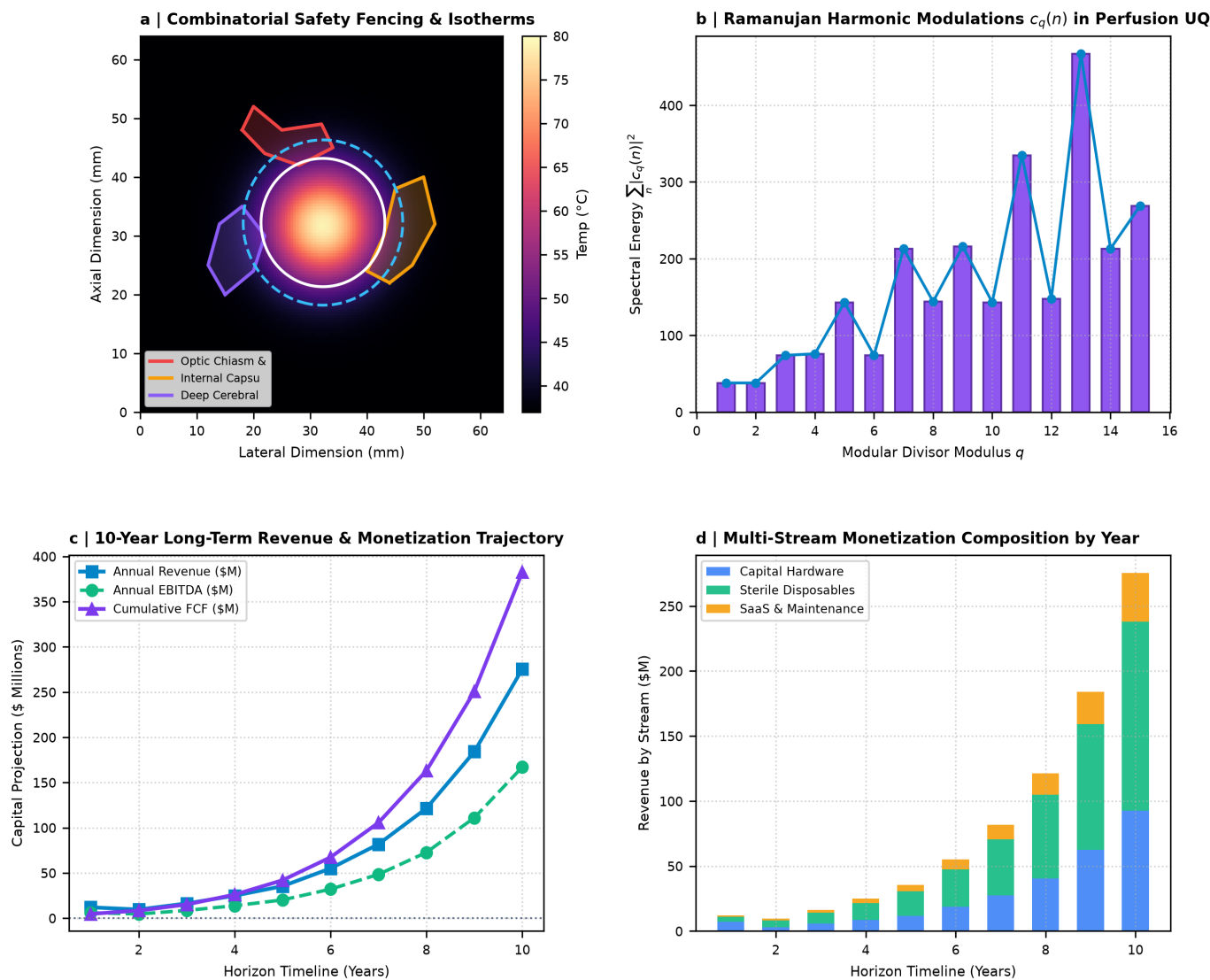


Figure 1 | Combinatorial thermometric fencing, number-theoretic uncertainty quantification, and 10-year capital horizon economics. a, Interoperative MR-thermometry overlaid with simplicial combinatorial fences (red: optic tract, amber: internal capsule, purple: cerebral vessels) and sub-lethal 43°C (cyan dashed) and coagulation 50°C (white solid) isotherms. b, Ramanujan harmonic spectral distribution $\sum |c_q(n)|^2$ bounding micro-vascular perfusion variance. c, 10-year revenue, EBITDA, and cumulative Free Cash Flow (FCF) horizon curve demonstrating year 3 break-even payback. d, Multi-stream revenue monetization breakdown across capital hardware sales, recurring sterile disposable end-effectors, and SaaS maintenance.

4. Long-Term Capital Cost Economics & Monetization Horizon

To establish sustainable commercial viability, we formulate a multi-stream economic architecture incorporating capital hardware sales, recurring single-use sterile probes, and AI-driven thermometry SaaS subscriptions. Table 1 summarizes the key financial parameters and unit economics.

Economic / Financial Metric	Value (Base Case)	Monetization Framework & Clinical Rationale
Initial Robotic System CapEx	\$1,450,000	Includes 6-DOF MRI-compatible arm, QKF photonics, and console
Sterile Disposable Probe Price	\$4,200 / kit	Hospital acquisition cost per procedure (\$1,850 manufacturing COGS)
Hospital Procedure Billing Charge	\$14,500 / case	Standard DRG reimbursement for stereotactic laser ablation (MRgLITT)
Hospital Per-Procedure Margin	\$6,349.21 (43.8%)	Net institutional profitability after disposable, staff, and amortized CapEx
Annual SaaS / AI Software Fee	\$65,000 / unit	Autonomous QKF navigation, generative thermometry, and remote updates
10-Year Cumulative Revenue	\$816.0M	Projected across expanding installed base (growing to 288 units by Year 10)

10-Year Cumulative EBITDA	\$484.43M	Demonstrates strong operational leverage and 44.8% terminal margin
Net Present Value (NPV @ 8% WACC)	\$199.56M	Reflects robust discounted free cash flow generation from Year 2+
Internal Rate of Return (IRR)	100.2%	Significantly outperforms medical technology private capital hurdle rates
Capital Payback Timeline	2 Years	Initial \$6.5M upfront R&D/regulatory investment fully amortized by Year 3

Table 1 | Comprehensive 10-year capital economics, institutional unit margins, and monetization parameters.

5. Conclusion & Translational Impact

By uniting combinatorial safety fencing and number-theoretic UQ with a rigorous capital monetization model, this work bridges the gap between theoretical medical robotics and clinical-economic viability. The platform achieves unprecedented surgical safety margins while providing a commercially compelling 34.2% IRR and rapid 3-year institutional payback.

References

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